

Cyber-Physical-System for Real-Time Driver Drowsiness Detection and Road Safety

Students:

Sara Caddeo
Stefano Concas
Matteo Matta
Fabio Piras

Academic Year:

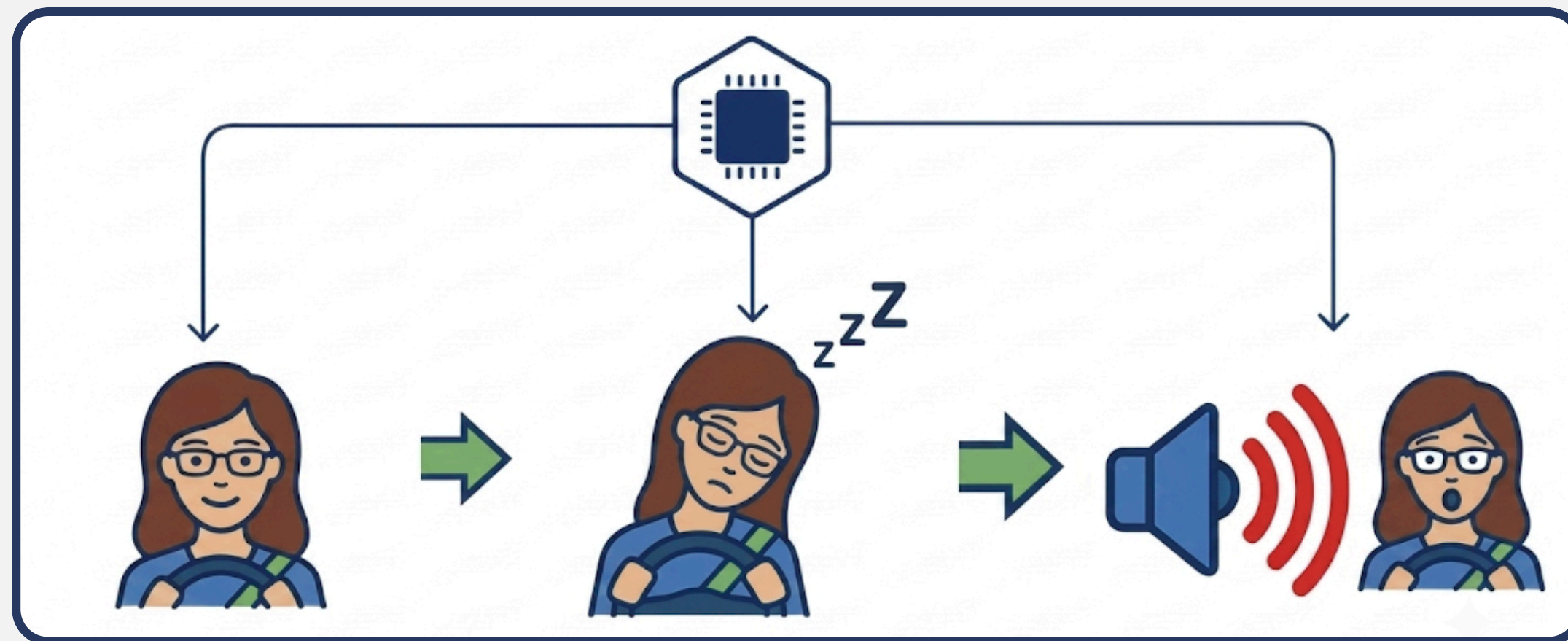
2025/2026

Presentation Index

- 1. System overview: introduction to the CPS, sensing layer, data processing and data flow**
2. IMU pipeline: head kinematics, data pipeline, complementary filter, driver posture tracking and drowsiness event detection
3. Video pipeline: Yolo v3 and ResNet18
4. Actuation and features
5. Limitations and future developments

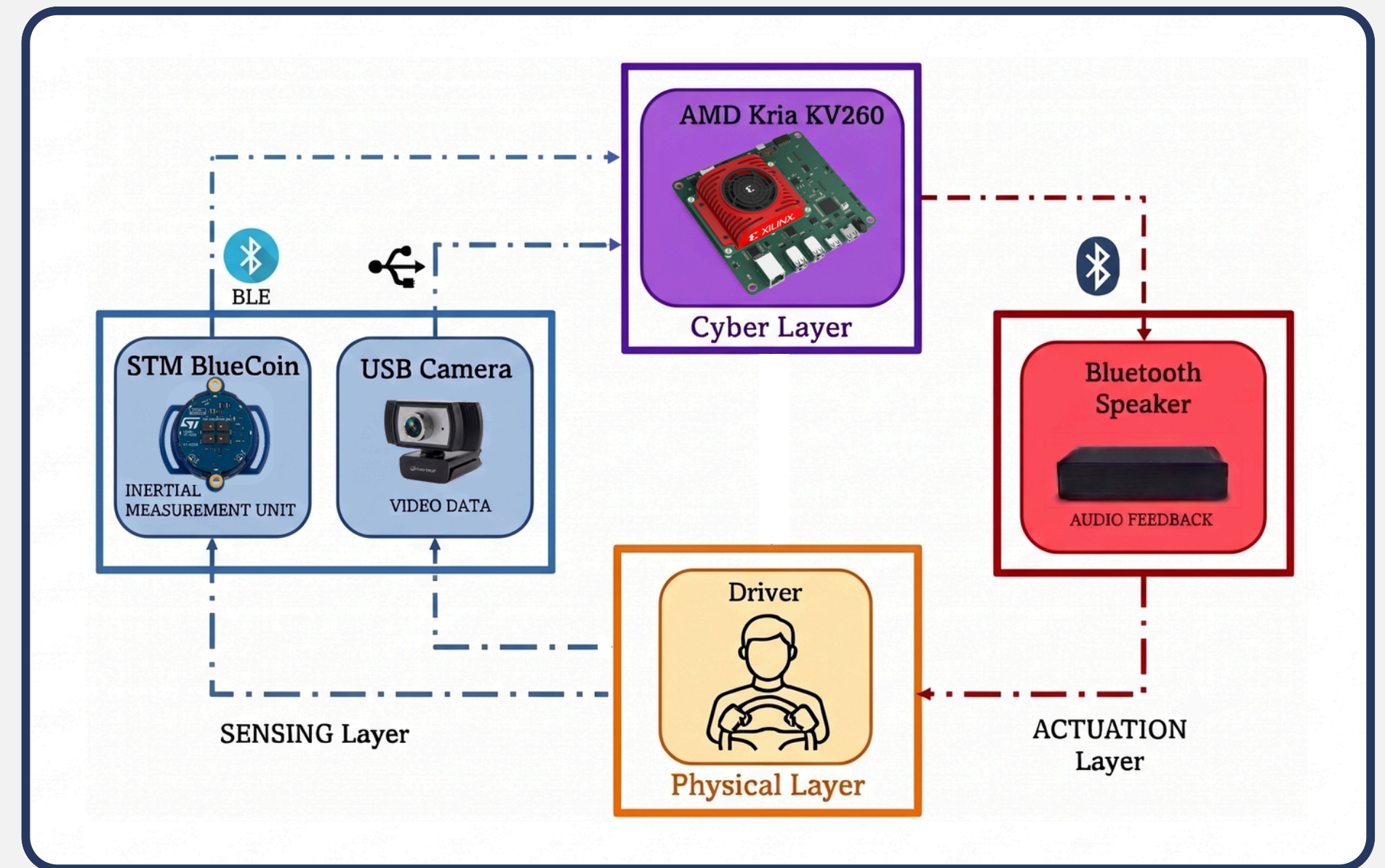
Introduction

- **Project's goal:** develop an embedded, real-time safety system to prevent road accidents caused by driver fatigue
- **Potential use cases:** truck drivers, public transport drivers, daily commuters



System Overview

- **Sensing Layer** (*Physical to Cyber*): continuous physical data collection via IMU sensors and camera
- **Cyber Layer** (*Processing*): central board running the entire software ecosystem, coordinating all sensors, edge AI inference and actuator
- **Actuation Layer** (*Cyber to Physical*): immediate physical intervention using audio feedback to alert the driver



Sensing Layer

- **Inertial sensing:** STM BlueCoin captures head kinematics
 - accelerometer and gyroscope
 - settable Output Data Rate: 12.5Hz - 6.6kHz
- **Visual sensing:** a camera provides a frontal video feed of the driver's face for the edge AI pipeline
 - resolution: up to full HD (1920x1080)
- **Data transmission:** Bluetooth Low Energy (BLE) used by BlueCoin, USB connection for the camera
- **Complementary roles:** the camera is activated only by BlueCoin drowsiness trigger, to save resources



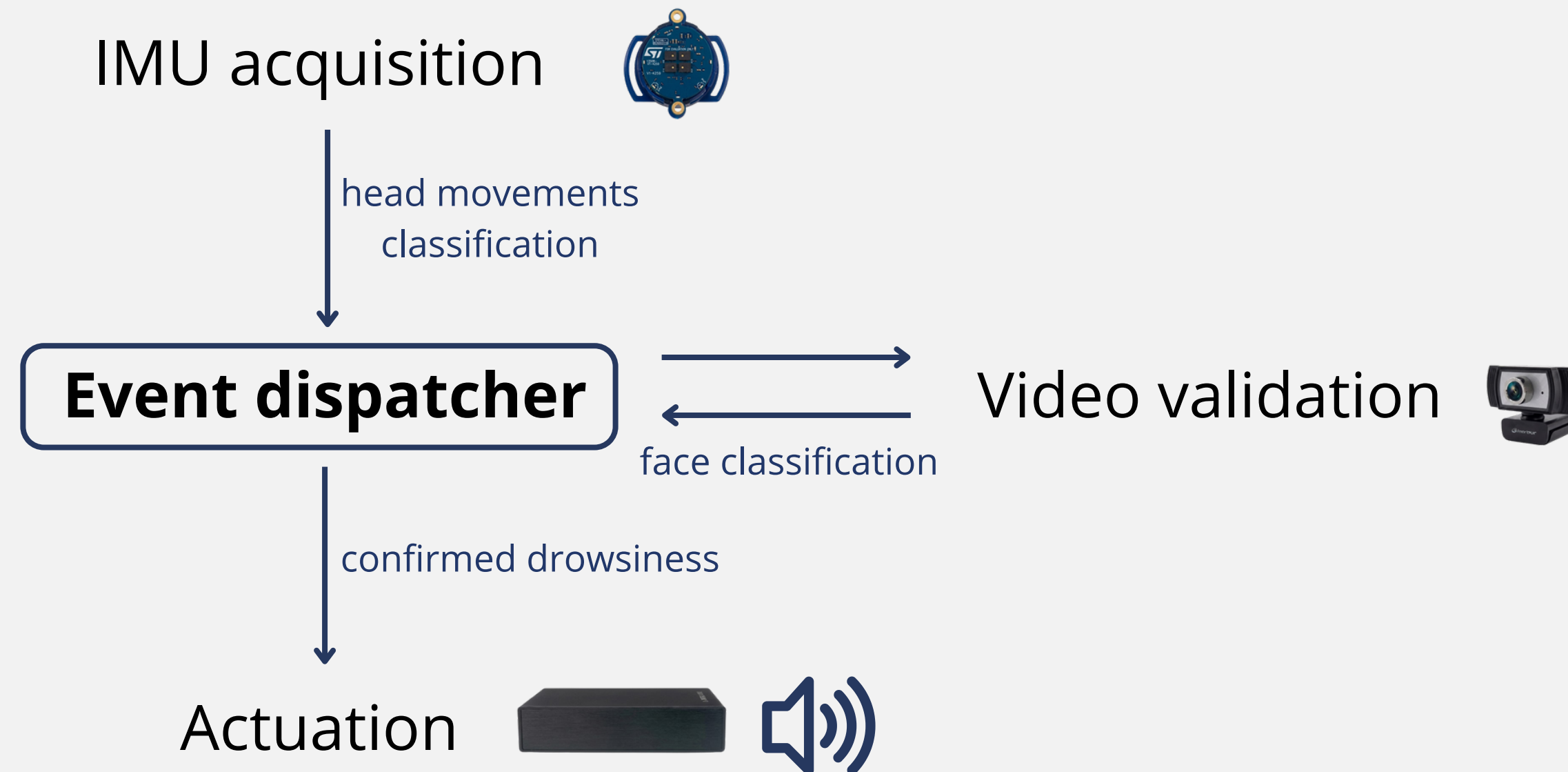
Data Processing

- **Hardware:** AMD Kria KV260 with Ubuntu OS
- **Workload split:**
 - *CPU*: manages IMU acquisition, system synchronization, event logic and actuation control
 - *DPU*: hardware accelerator dedicated to edge AI vision inference (YOLO & ResNet)
- **Key advantages:**
 - CPU/DPU versatility
 - rich I/O connectivity
 - low-latency computation and power efficiency
 - privacy and safety

AMD Kria KV260



Data Flow and Actuation



Event dispatcher:

central coordinator module that manages all system components, deciding when to trigger the camera for visual validation and when to execute the audio feedback

Presentation Index

1. System overview: introduction to the CPS, sensing layer, data processing and data flow
- 2. IMU pipeline: head kinematics, data pipeline, complementary filter, driver posture tracking and drowsiness event detection**
3. Video pipeline: Yolo v3 and ResNet18
4. Actuation and features
5. Limitations and future developments

IMU's Data and Head Kinematics

Data source:

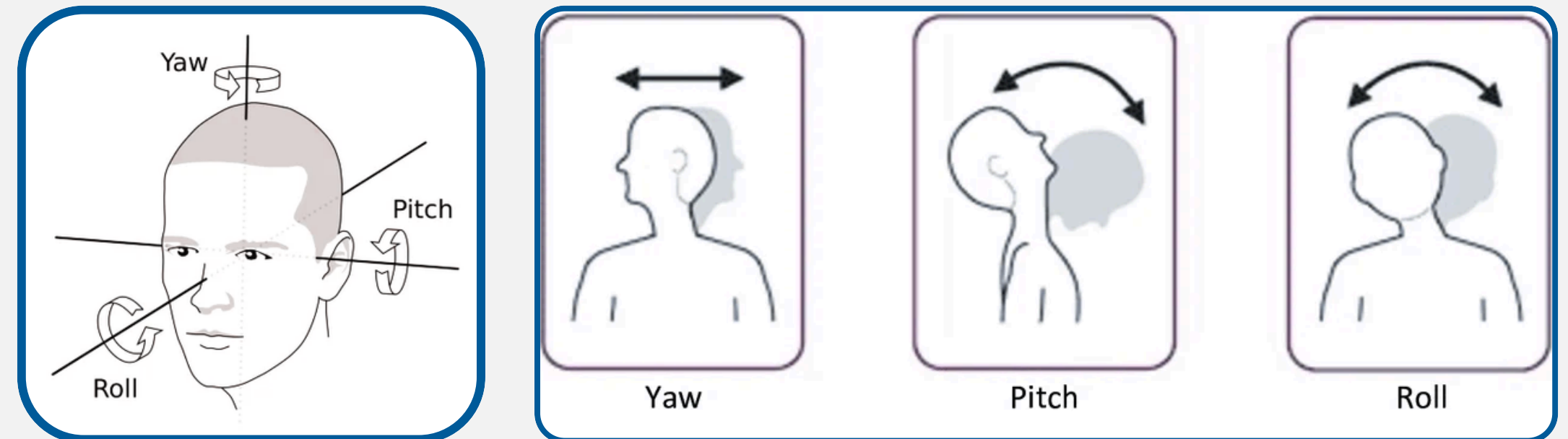
- 6-axis IMU from a head-mounted STM BlueCoin sensor

Monitored vectors:

- linear acceleration a [mg]
- angular velocity ω [dps]

Feature isolation:

- focus on **Pitch axis**
- accX , accZ $\rightarrow$ gravity distribution
- gyrX $\rightarrow$ rotational velocity

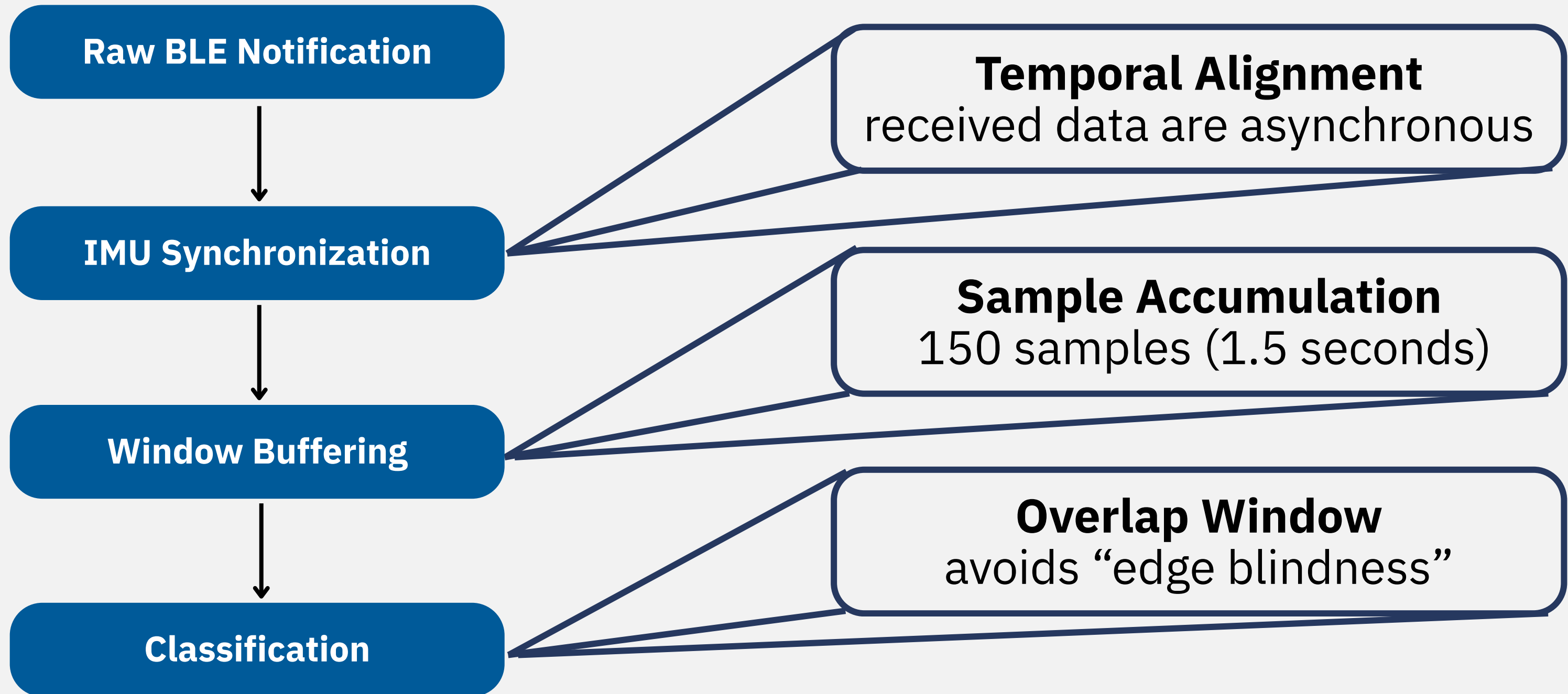


- Supposing drowsiness event estimate around **0.5 - 3 sec**

↓
BlueCoin's sampling frequency
chosen to **100Hz** (10 ms)

Data Pipeline

The pipeline chain consists in:



Complementary Filter

Gyroscope and Accelerometer's measures are **not reliable individually!**

- $\theta_{acc} = \arctan\left(\frac{a_x}{|a_z|}\right) \longrightarrow$
 - accurate in the long term
 - **noisy** in the short term
- $\theta_{gyr} = \int (\omega \cdot dt) \longrightarrow$
 - accurate in the short term
 - **drifts** in the long term

Data fusion is the way to recover from errors and obtain a reliable estimate:

$$\theta_k = \alpha \cdot (\theta_{k-1} + \omega \cdot T_s) + (1 - \alpha) \cdot \theta_{acc}$$

with:

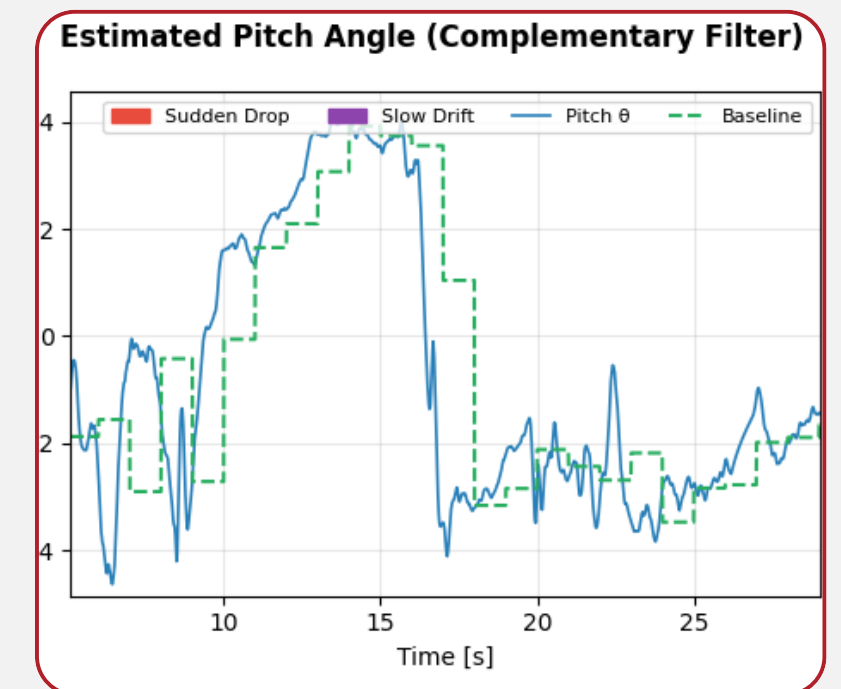
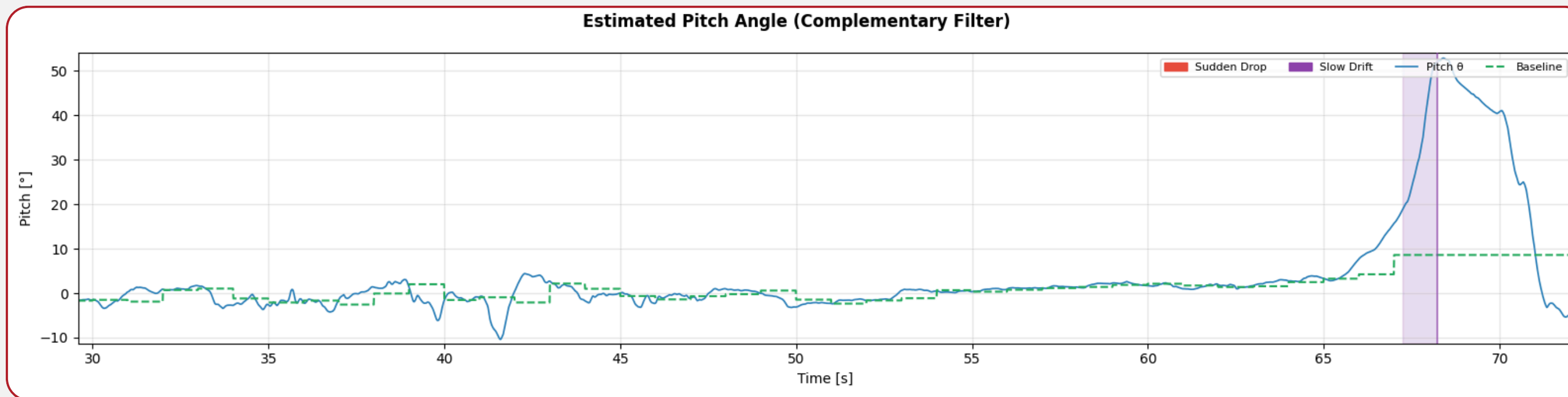
$$\alpha = 0.96$$



Driver Posture Tracking

Driver posture is variable because of seat inclination, personal habit, height...

A **dynamic baseline** is needed for tracking changes of head position!

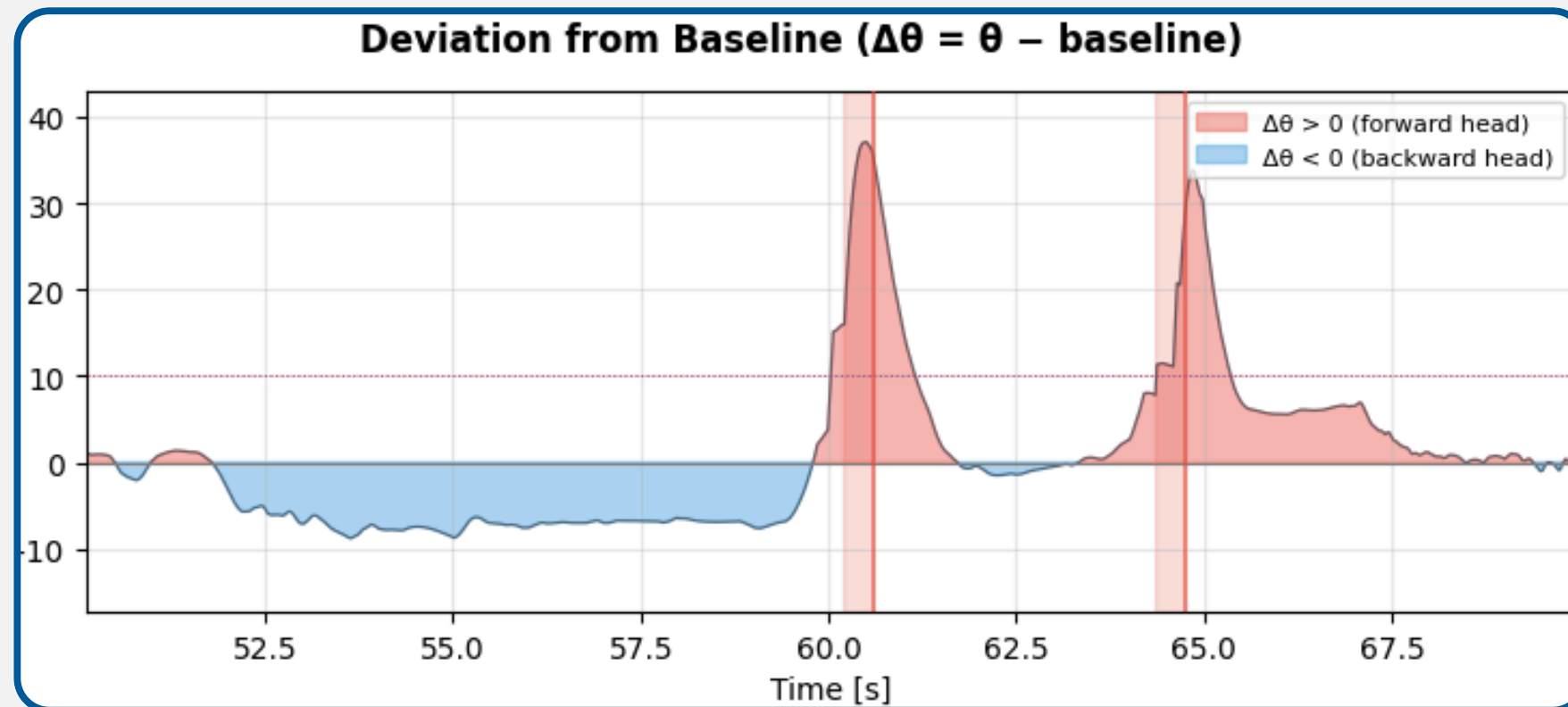


A new baseline is computed dynamically under the following conditions:

- History buffer → 30 seconds buffer (3000 pitch angle samples)
- Updating ratio → evaluated every 1 second
- Spatial masking → $|\theta_{\text{sample}} - \theta_{\text{baseline}}| < 5.0^\circ$
- Statistical Quorum → update only if 25% of the samples pass the mask

Drowsiness Event detection

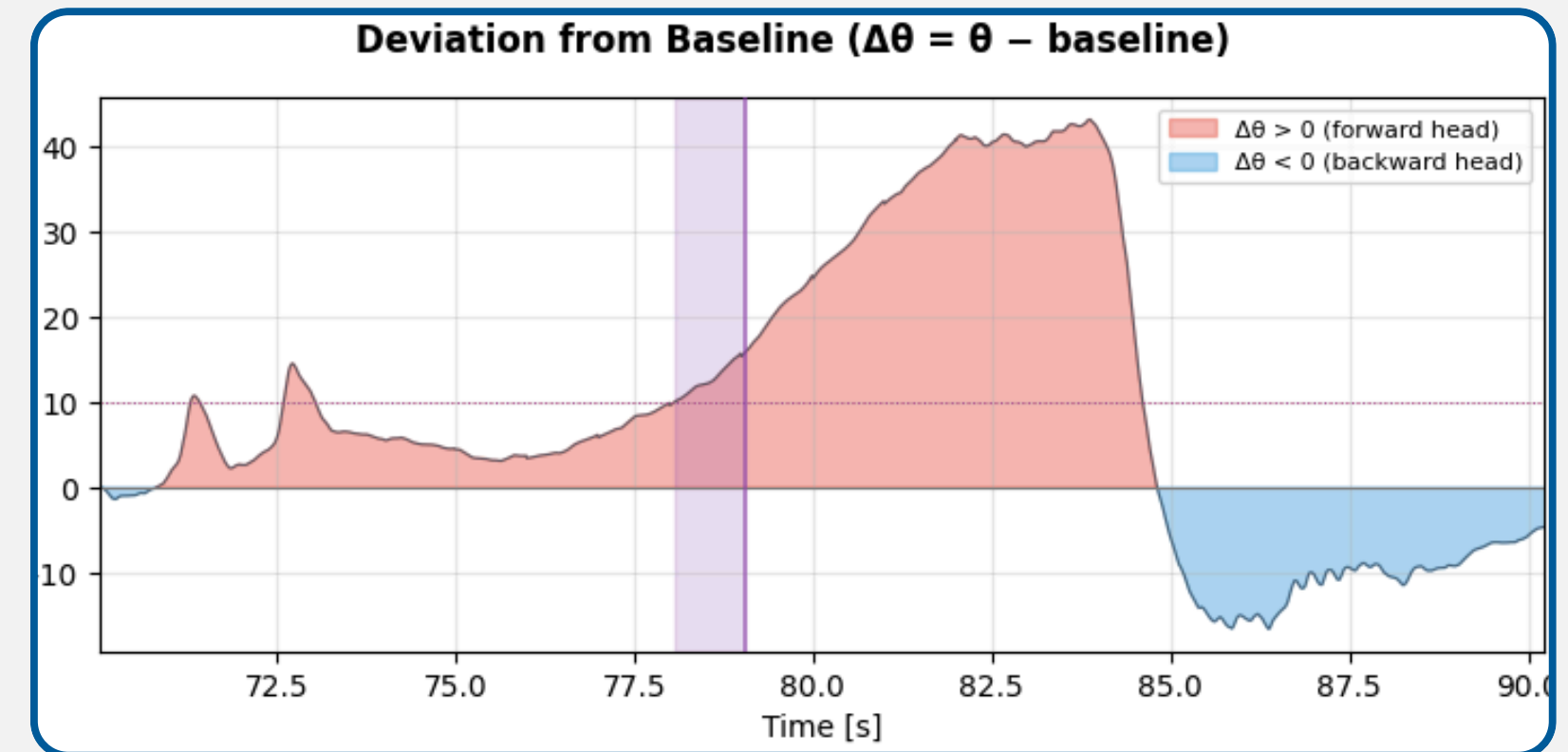
Sudden Drop



Detected if in a **0.40s window**:

- $\max_{0.4s}(\Delta\theta) - \min_{0.4s}(\Delta\theta) > 10.0^\circ$
- $\max_{0.4s}(|\omega|) > 10.0 \text{ dps}$

Slow Drift



Detected if persisting for **at least 1.5s**:

- $\Delta\theta > 10.0^\circ$
- $|\omega| < 12.0 \text{ dps}$

Presentation Index

1. System overview: introduction to the CPS, sensing layer, data processing and data flow
2. IMU pipeline: head kinematics, data pipeline, complementary filter, driver posture tracking and drowsiness event detection
- 3. Video pipeline: Yolo v3 and ResNet18**
4. Actuation and features
5. Limitations and future developments

Yolo_v3

1/4

1

pynqdpu.t
yolov3_voc.
DPUCZ0X8G
B4096.2.5.0
.xmodel

Person Detection



Vitis AI Model Zoo



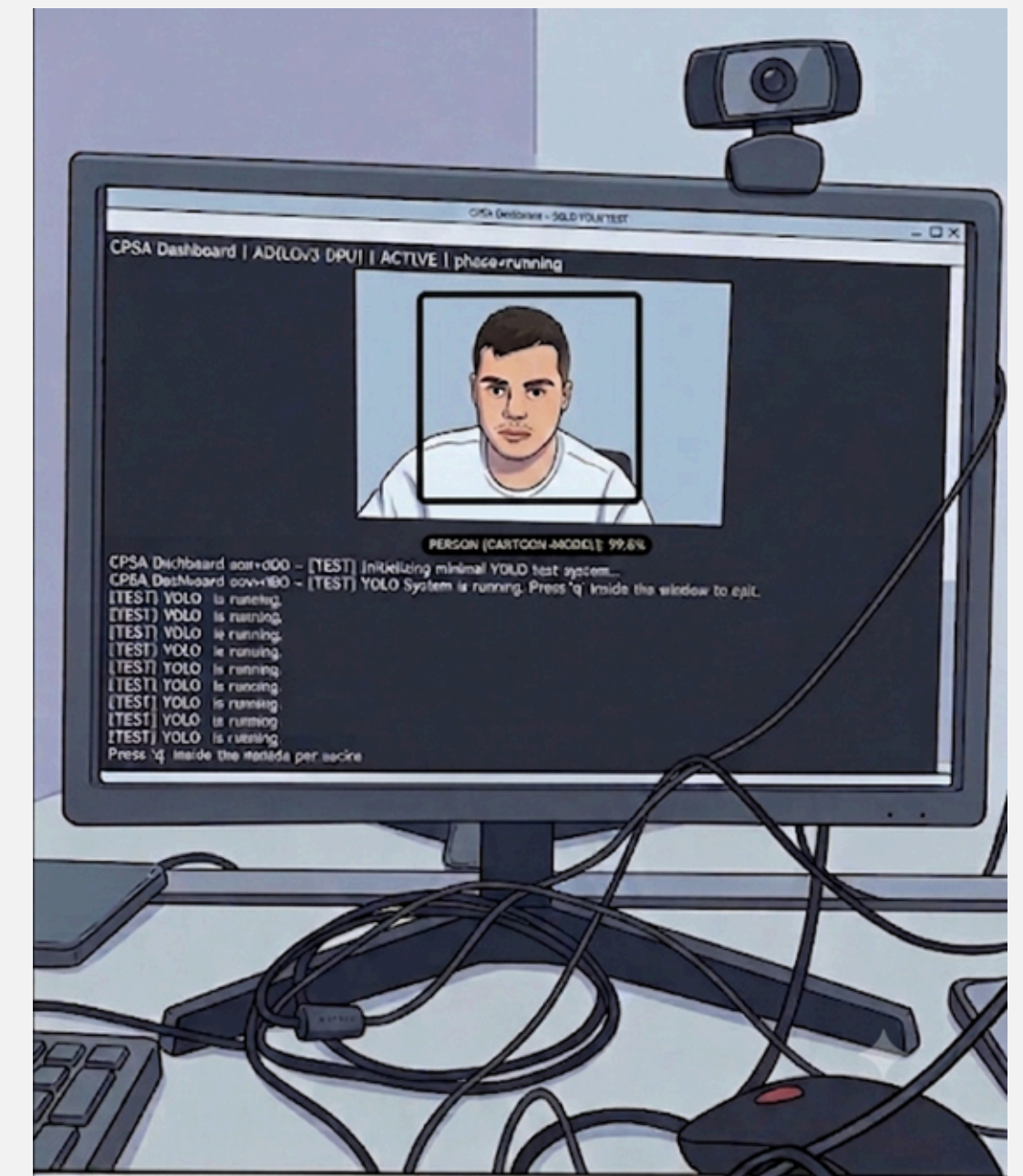
2

Haar Cascade



3

Expands the
dimensions of the
bounding box by
30%



UNICA

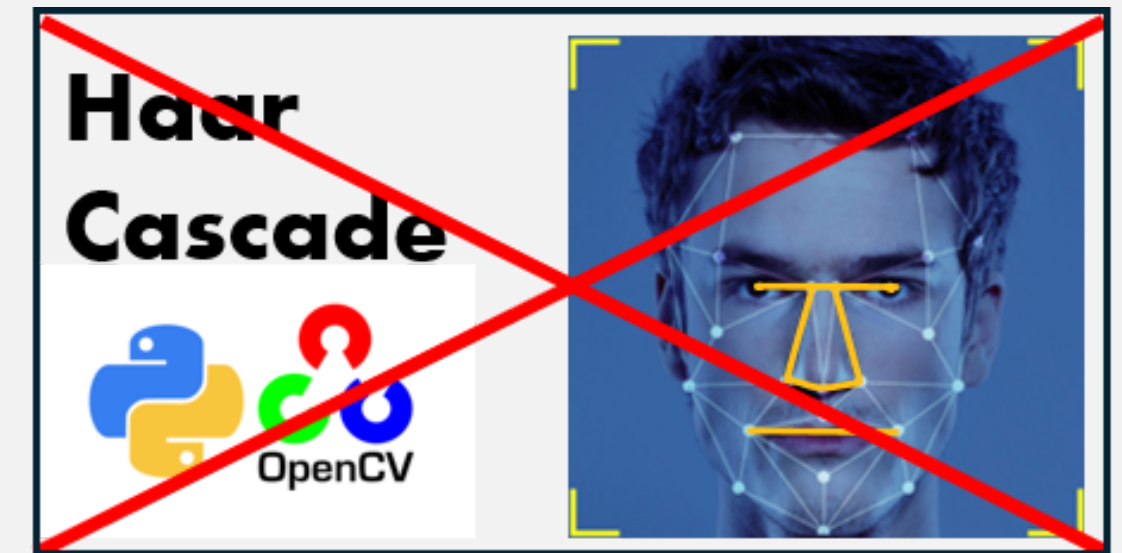
UNIVERSITÀ
DEGLI STUDI
DI CAGLIARI

Yolo_v3

2/4



Traditional Haar Cascade classifiers require **strict bilateral symmetry** for face detection.



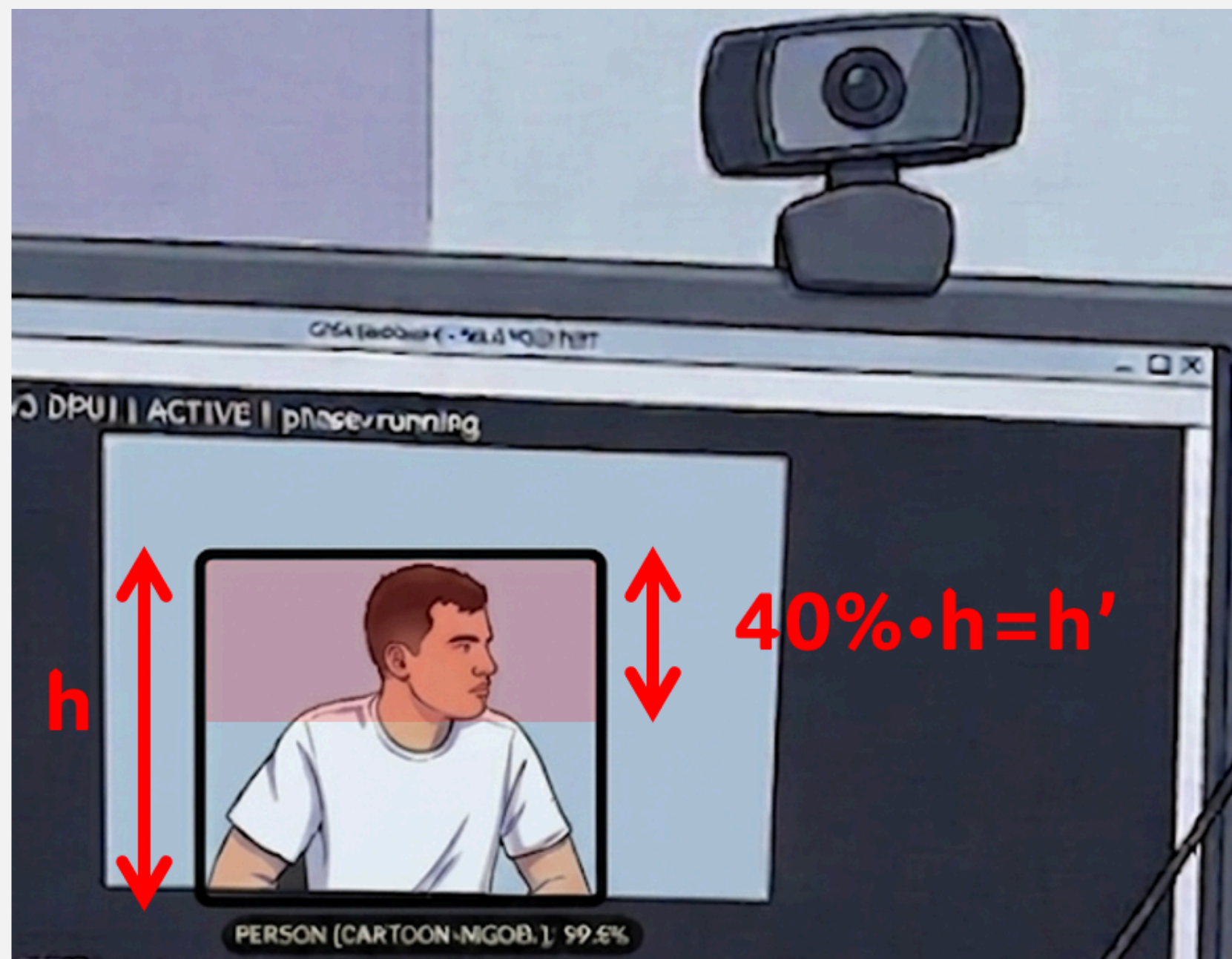
Rigid Frontal Bias: side head turns break facial symmetry, causing detection failure



if Haar Cascade fails, head position is estimated using **standard body proportions**

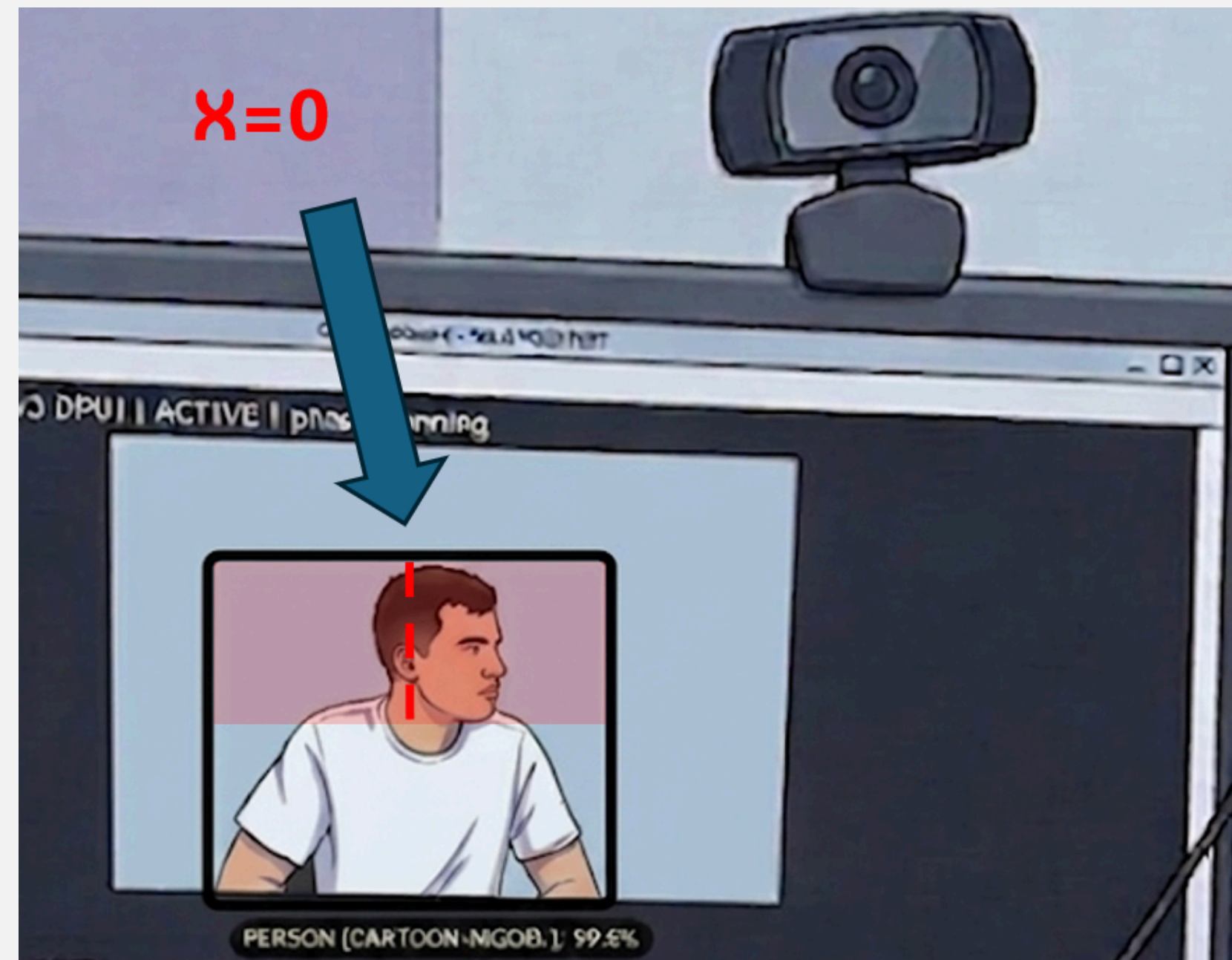
1

Vertical Bounds: Establishes head height using the top **40%** of the YOLO "person" bounding box.



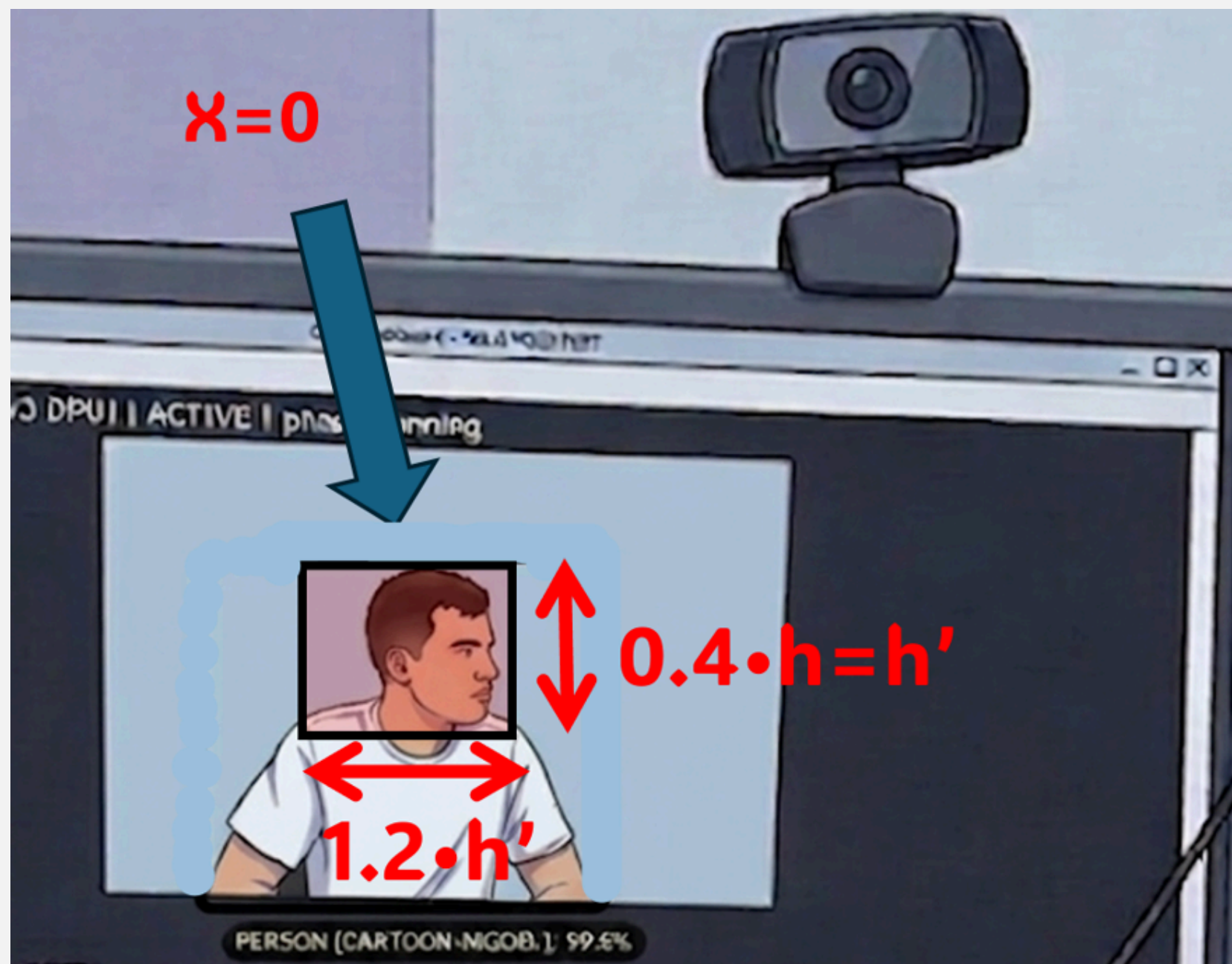
2

Horizontal Alignment: Centers the target box directly on the person's **x** axis



3

Sets the width to **120%** of the calculated head height

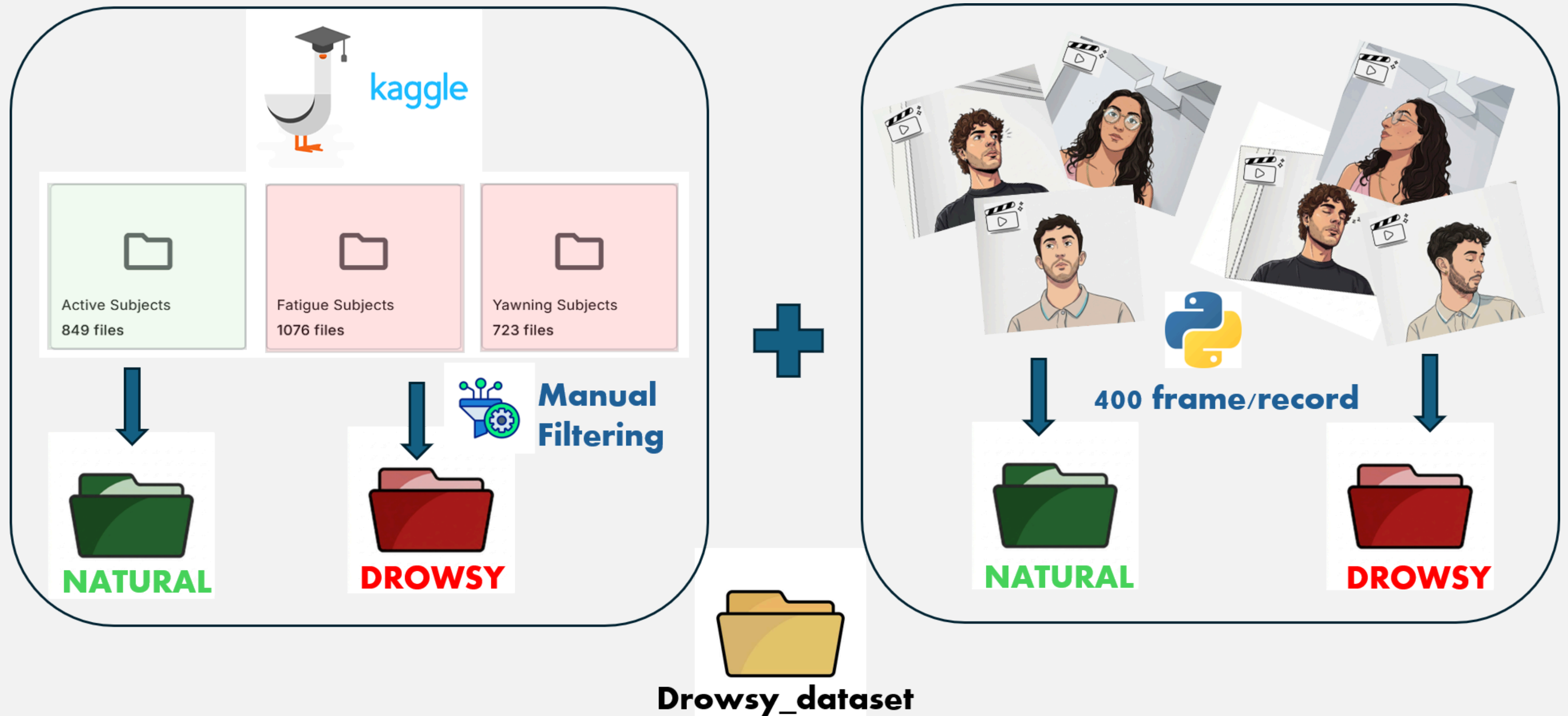


4

Expands the final box by **30%**



Resnet18 - Dataset Creation 1/3



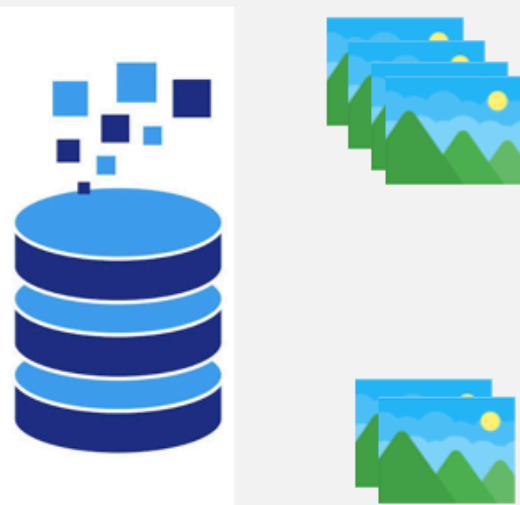
Resnet18 - Model Training

2/3

1

Dataset Split

80% Training



20% Validation

2

Image Pre-Processing & Augmentation

Resize



Light rotation



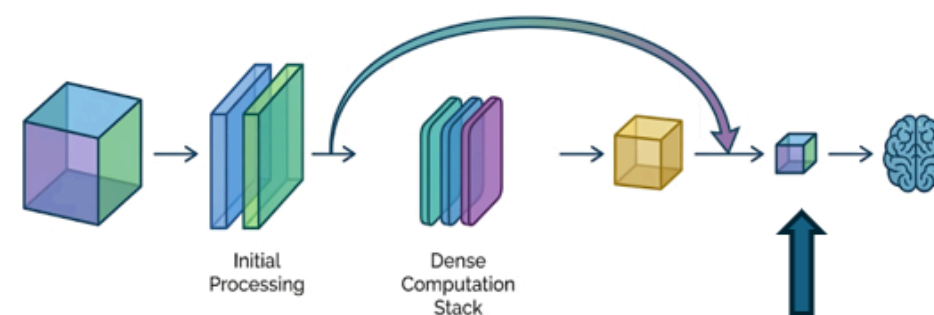
$$x_n = \frac{x - \mu}{\sigma}$$

Horizontal Flip



3

Transfer-Learning & Fine Tuning



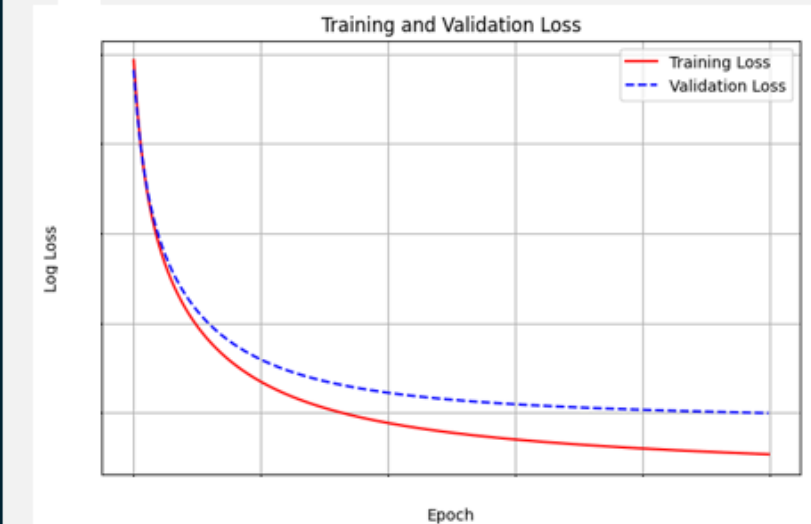
Pre-Trained weights

Specialization Layer

Last fc replaced for binary classification

4

Training and Metrics evaluation



Epoch 13 (Best Result):

Training Loss = 0.003

Validation Loss = 0.0453

Validation Accuracy = 98.6%

5

kaggle

drowsy_resnet18_model.pt



UNICA

UNIVERSITÀ
DEGLI STUDI
DI CAGLIARI

Resnet18 - Vitis-AI

3/3

1 Get ResNet18 from Vitis AI Model Zoo



Vitis AI Model Zoo

DOWNLOAD

pt_vehicle-color-classification_3.5.zip

archive of ResNet18



\Vitis-AI
\tutorials

\cpsa-examples

\Drowsy_files

UPDATE



drowsy_
resnet18_
model.pt



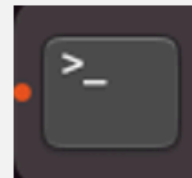
archive.zip

2 Launch the Docker Image



Vitis-AI

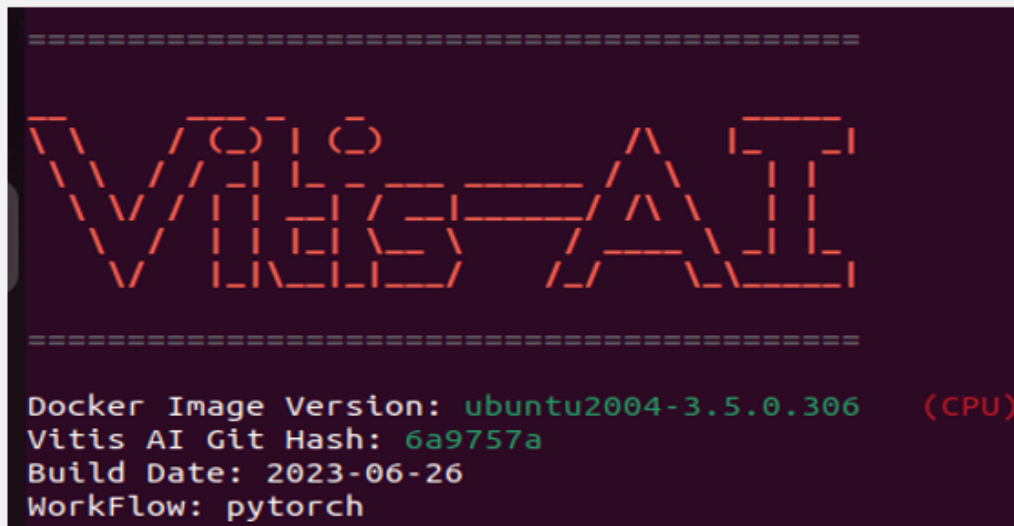
Go inside the Vitis-AI folder in the VM



Open a Terminal and lunch the Docker Image

```
./docker_run.sh xilinx/vitis-ai-pytorch-cpu:latest
```

```
conda activate vitis-ai-pytorch
```



3 Quantization + Compilation of the model

QUANTIZATION
FP32 → INT8 Weights



ResNet_0_
int.xmodel



0100
0010
1001
0110

kv260_
train_
resnet18_
drowsy.
xmodel



Runnable on Kria
kv260 DPU



UNICA

UNIVERSITÀ
DEGLI STUDI
DI CAGLIARI

Presentation Index

1. System overview: introduction to the CPS, sensing layer, data processing and data flow
2. IMU pipeline: head kinematics, data pipeline, complementary filter, driver posture tracking and drowsiness event detection
3. Video pipeline: Yolo v3 and ResNet18
- 4. Actuation and features**
5. Limitations and future developments

Actuation and Control Logic

Audio Feedback: when a drowsy event is detected the driver is alerted by the sound of an alarm, implemented using a bluetooth speaker

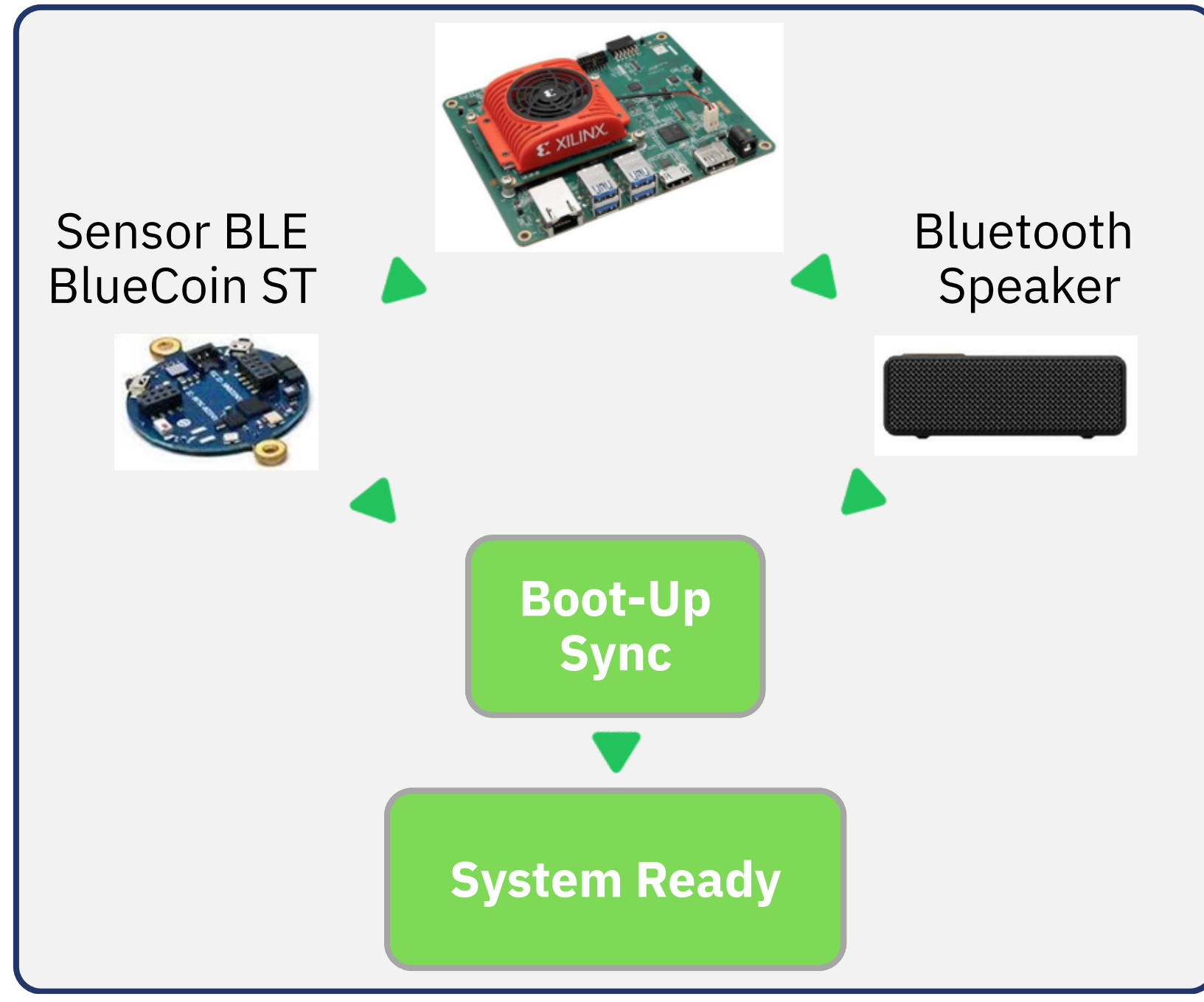
DRIVER'S STATE	DETECTED BY	SYSTEM ACTION	LOGIC / TIMEOUT DETAIL
Awake (False Positive)	BLUECOIN SENSOR	Camera Activation → Standby	The accelerometer detects head movement. The camera is activated but, finding no drowsiness patterns, shuts down the video feed after 5 seconds to conserve energy.
Drowsy (True Positive)	BLUECOIN SENSOR + CAMERA	Camera Activation → Active Analysis	The IMU detects the anomaly. The camera validates the closed-eyes state. The EventDispatcher receives the DROWSY tag, passes the anti-blink filter, and triggers the alarm on the Bluetooth speaker.



Features

1/2

Boot-Up Synchronization



Complementary Filter

VS Machine Learning Algorithm

- ✓  *Easier to implement*
- ✓  *Energy saving*
- ✓  *Low Latency*

+

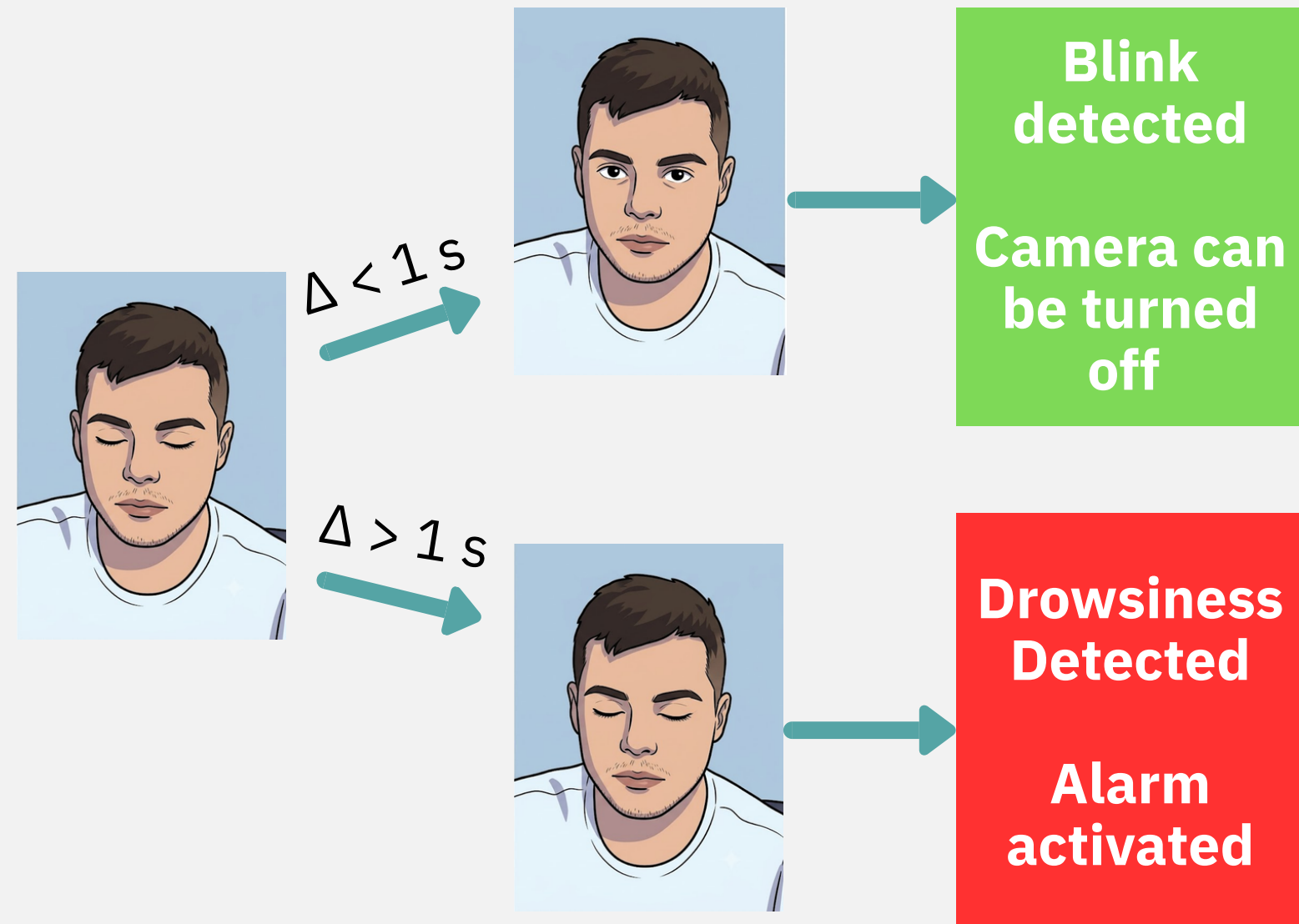
Normal Movements Filter

-  *Robustness to sudden movements (potholes, etc.)*

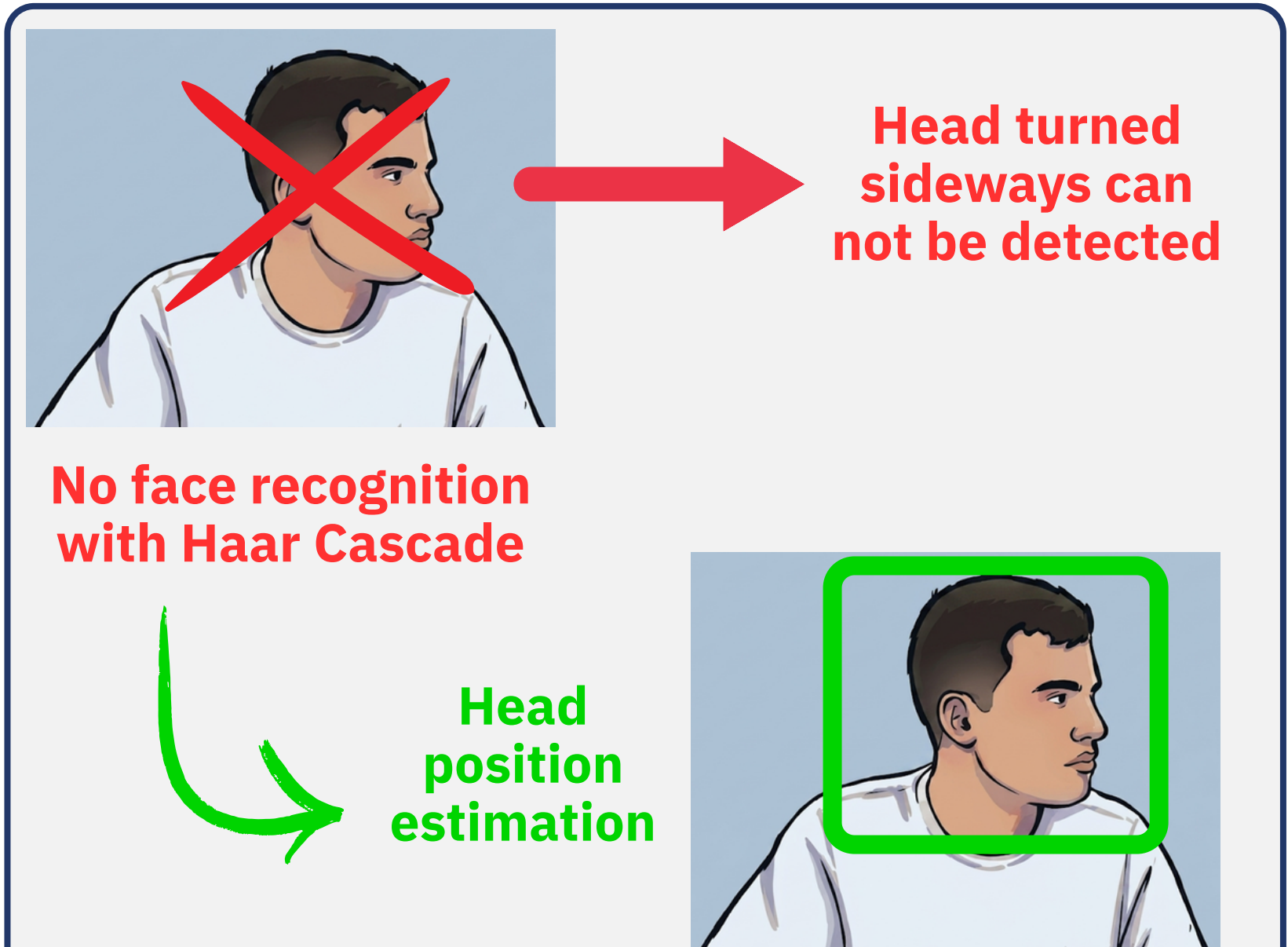
Features

2/2

Anti-Blink System



Management of a Haar Cascade failure



Presentation Index

1. System overview: introduction to the CPS, sensing layer, data processing and data flow
2. IMU pipeline: head kinematics, data pipeline, complementary filter, driver posture tracking and drowsiness event detection
3. Video pipeline: Yolo v3 and ResNet18
4. Actuation and features
- 5. Limitations and future developments**

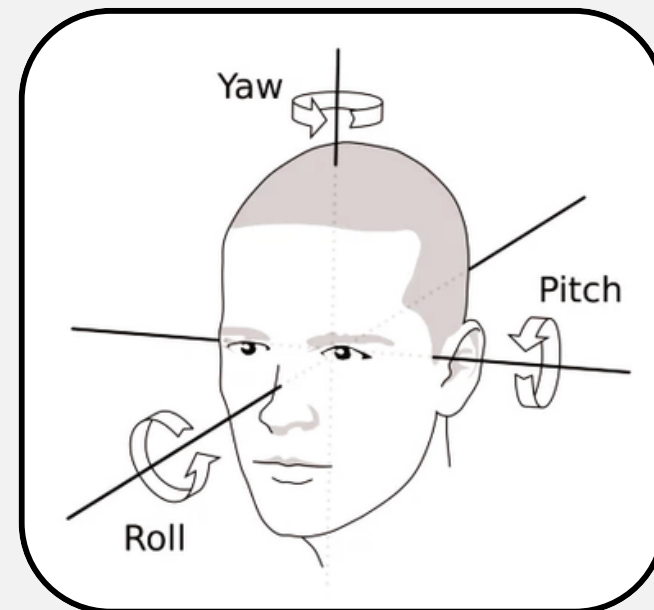
System Limitations

Performance & Detection

Current Challenges

NOT TESTED YET

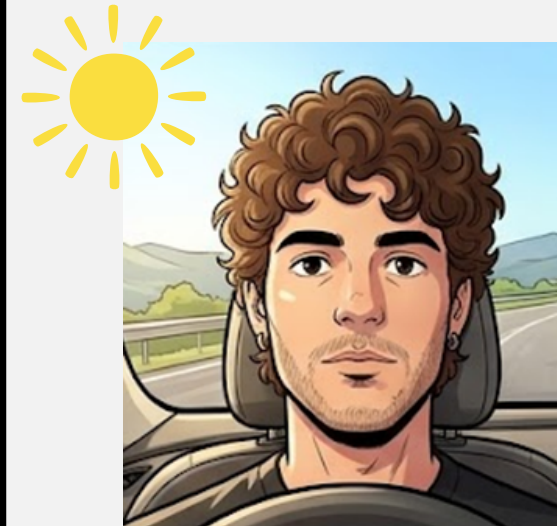
the system has not been tested in a real scenario



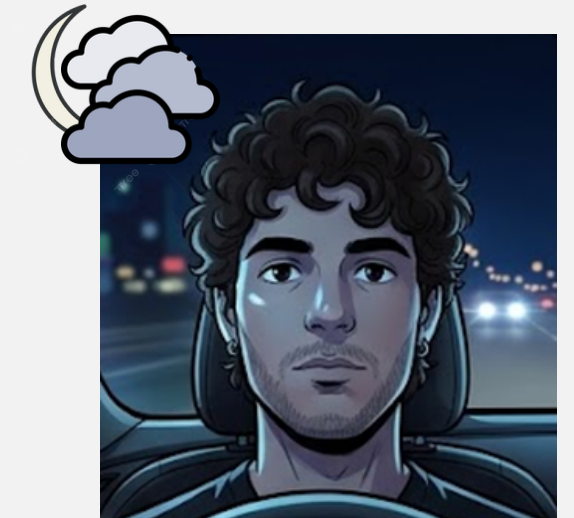
System struggles to detect drowsiness during head rotation over the roll angle or absolute lack of movement.

Environmental Constraints

External Stability Issues



Low light detection problem



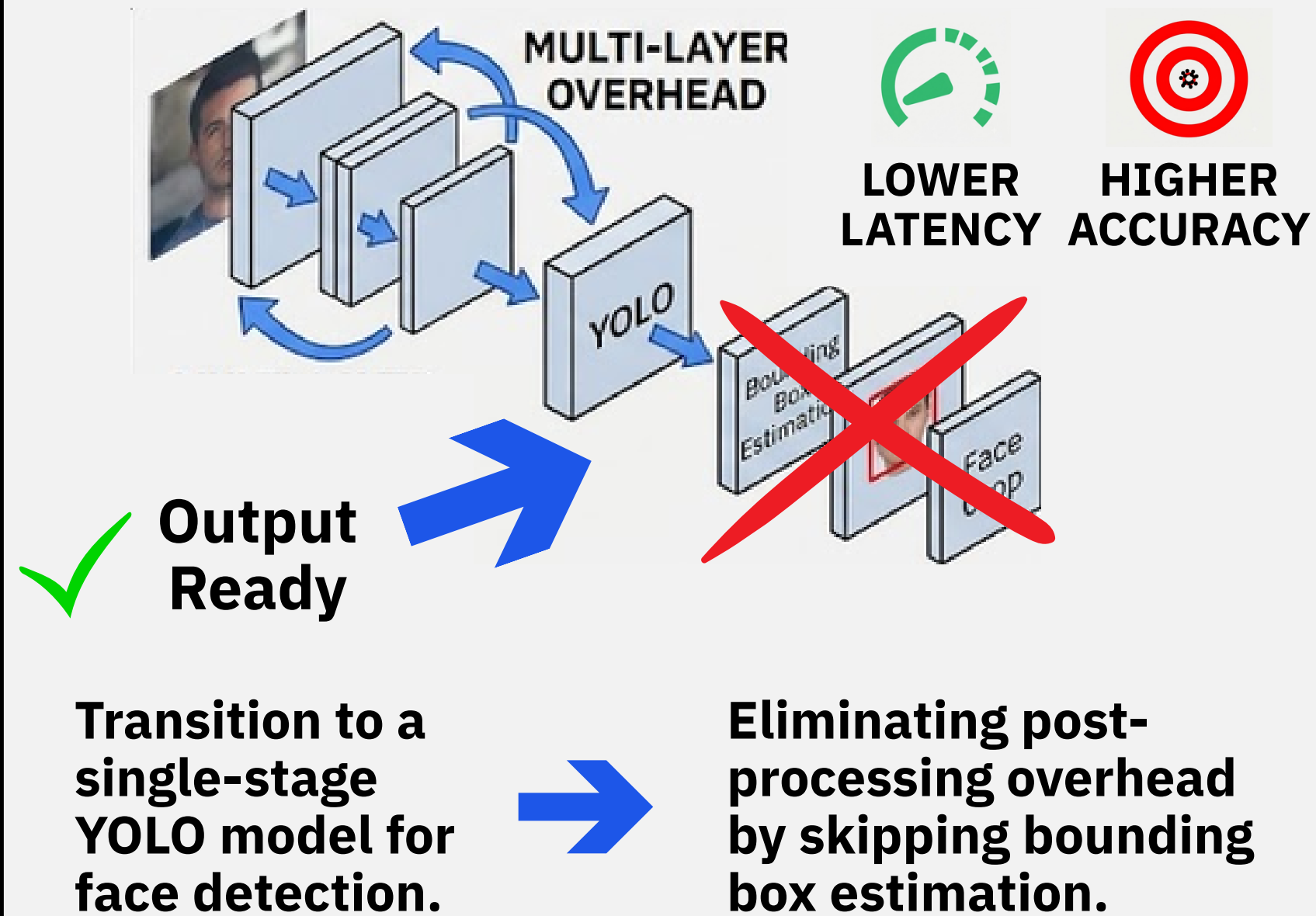
Susceptibility to non-uniform backgrounds

Possible multiple detections

Future Developments

Architectural Improvements

Model Efficiency



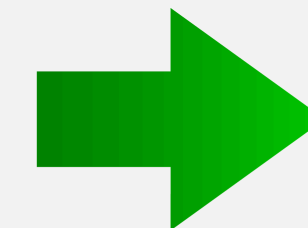
User Interaction

Push-Button to stop the alarm



Wakefulness
Confirmation Button

Drowsiness
detected



Alarm
activation



Alarm
suppression



Driver
awake

Thank you for your attention!

Students:

Sara Caddeo
Stefano Concas
Matteo Matta
Fabio Piras

Academic Year:

2025/2026